Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	24 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to athletes.csv, games_summary.csv, And alltime_investment.csv flat files.	High
2	Data Processing	Create relationships in Tableau using noc_code and year as join keys	High
3	Business Analysis	Calculate Average Medals per Athlete, Medal Efficiency Percentage, and Dynamic KPI selection.	High
4	Business Insights & Analytical Outputs	Provide 8 detailed visualizations including Maps, Line Charts, Tree Maps, and Bubble Charts.	High
5	User Interaction & Analysis	Implement dropdown interactive filters for Country, Region, Year, and Sport Category.	High
6	Data Update & Reporting	Publish the interactive Story to Tableau Public and embed it into a Flask web application.	High

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Flask frontend must use a responsive Bootstrap template (Arsha) for mobile and desktop viewing.
NFR-2	**Security**	Tableau dashboards must be embedded in read-only mode to prevent unintended data alterations.
NFR-3	**Reliability**	The dashboard should handle all 8,000+ rows of athlete data without crashing or stalling.
NFR-4	**Performance**	Tableau's in-memory engine must execute cross-calculations and filter updates in under 3 seconds.
NFR-5	**Availability**	The visualizations must be hosted on Tableau Public ensuring 24/7 uptime for the web application.
NFR-6	**Scalability**	The data model should easily accept updated CSV files for the 2028 Olympics without requiring a full redesign.